

RES=RAG and the Coherent State Network Protocol

A Formal Architecture for Endogenous Stability in Human-Machine Systems

Canonical formalization and deterministic reference profile, version 1.0.0

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Epistemic status. This is a formal specification and testable research programme. It does not report a completed empirical validation, prove machine consciousness, or claim that human-machine interaction is a physical quantum system.

Abstract

RES=RAG models stable intelligence as a regulated tension between receptive, context-grounded dynamics (RES) and internally generative, novelty-producing dynamics (RAG). The central claim is not that the two distributions become equal. Stability instead occupies a calibrated, non-zero optimal-transport band: too little separation produces rigidity, while excessive separation produces instability, fabrication, or loss of coordination.

This paper consolidates the conceptual framework introduced by Jean-Charles Tassan with the optimal-transport, information-substrate, operational-variable, and systems-validation work developed with Trent Slade. It supplies a common state space, measurable quantities, explicit failure conditions, and a protocol boundary. The Coherent State Network Protocol (CSNP) turns the model into a cyclic observation-classification-intervention-receipt process. A deterministic reference profile, CSNP-RP 1.0.0, defines canonical JSON receipts, hash chaining, calibration declarations, evidence references, and replay rules.

The result is deliberately modest in epistemic scope. RES=RAG is presented as a falsifiable architecture for studying regulated dialogue and distributed human-machine coordination. Thresholds are calibration parameters, not universal constants; Wasserstein distances depend on declared encoders and sampling procedures. The contribution is therefore a reproducible theory and protocol specification from which empirical studies can be designed.

1 Scope and provenance

RES=RAG begins from a relational thesis: stable cognition is not adequately described by reception or generation alone, but by their regulated co-presence across human, machine, and interaction domains. The originating framework and three-axis organization are due to Tassan. Subsequent joint work introduced an optimal-transport reading, a history-dependent informational substrate, dual operators, and interface-level observables [Tassan and Slade, 2025, Slade and Tassan, 2025, 2026]. The canonical repository then separated axioms, predictions, falsifiers, and ethical constraints [Tassan and Slade, 2026].

This version has four goals:

1. preserve the canonical distinction between receptive and generative dynamics;
2. express equilibrium as a bounded, non-zero transport relation;

3. connect the theory to observations available at a human-machine interface; and
4. define an auditable protocol and deterministic receipt format.

Only work directly connected to RES=RAG is included. Broader field theories, unrelated QSOL-IMC systems, and uncited external projects are outside scope.

1.1 Claim labels

Label	Meaning in this paper
Definition	A convention that gives the specification a precise meaning.
Protocol rule	A normative requirement for CSNP-RP conformance.
Hypothesis	A claim that can be tested and rejected.
Reported target	A value from exploratory work that is not established here.
Limitation	A known boundary on interpretation or implementation.

Table 1: Epistemic labels used throughout the formalization.

Claims such as a 40-fold memory reduction, hallucination below 0.02 percent, coherence near 0.96, or universal collapse thresholds are not results of this formalization. If adopted by an implementation, they must be marked as external, unvalidated targets and coupled to a citable measurement procedure.

2 Formal objects

Let $(\mathcal{S}, d)$ be a declared Polish metric space, and let $\mathcal{P}_2(\mathcal{S})$ be the probability measures on $\mathcal{S}$ with finite second moment. A state can represent a semantic embedding, dialogue state, task state, or another reproducible encoding. The representation, sampling rule, and ground metric are part of the study specification.

Definition 1 (Functional distributions). *At observation t ,*

$$\mu_t^{\text{RES}} \in \mathcal{P}_2(\mathcal{S}) \quad \text{and} \quad \mu_t^{\text{RAG}} \in \mathcal{P}_2(\mathcal{S})$$

denote respectively the receptive, contextual, evidence-conditioned distribution and the generative, novelty-producing distribution.

These are functional roles. RAG here means regenerative or generative dynamics; it is not restricted to the engineering pattern usually called retrieval-augmented generation.

For $\mu, \nu \in \mathcal{P}_2(\mathcal{S})$, the quadratic Wasserstein distance is

$$W_2^2(\mu, \nu) = \inf_{\gamma \in \Pi(\mu, \nu)} \int_{\mathcal{S} \times \mathcal{S}} d(x, y)^2 d\gamma(x, y), \quad (1)$$

where $\Pi(\mu, \nu)$ is the set of couplings with marginals μ and ν [Villani, 2009, Ambrosio et al., 2008]. Define

$$w_t = W_2(\mu_t^{\text{RES}}, \mu_t^{\text{RAG}}). \quad (2)$$

2.1 Bounded equilibrium

Definition 2 (RES=RAG equilibrium). *For calibration profile $\theta = (\varepsilon_{\min}, \varepsilon_{\max}, \dots)$, where $0 \leq \varepsilon_{\min} < \varepsilon_{\max}$, the stable set is*

$$\mathcal{E}_\theta = \left\{ (\mu^{\text{RES}}, \mu^{\text{RAG}}) : \varepsilon_{\min} \leq W_2(\mu^{\text{RES}}, \mu^{\text{RAG}}) \leq \varepsilon_{\max} \right\}. \quad (3)$$

The equality sign in RES=RAG names this regulated relation. It does not require pointwise identity or zero transport. Below the lower boundary, the process risks sterility, repetition, or over-constraint; above the upper boundary, it risks drift, fabrication, or loss of shared reference.

The distance-to-band diagnostic

$$e_\theta(w_t) = \max(0, \varepsilon_{\min} - w_t) + \max(0, w_t - \varepsilon_{\max}) \quad (4)$$

is non-negative and vanishes precisely inside the declared stable interval.

Proposition 1 (Non-minimization). *Minimizing w_t without the lower bound is not a valid RES=RAG objective.*

Proof. An unconstrained minimum admits $w_t = 0$. By [equation \(3\)](#), that point is outside the stable set whenever $\varepsilon_{\min} > 0$, and it erases the productive difference the model is intended to preserve. $\square$

2.2 Temporal-reference discrepancy

Let μ_t^H and μ_t^M be declared human-side and machine-side state estimates.

Definition 3 (Operational real-time discrepancy).

$$T_{\text{Real}}(t) = W_2(\mu_t^H, \mu_t^M). \quad (5)$$

T_{Real} is a discrepancy between situated reference frames. It is not physical clock time. A conforming empirical study must declare the estimator, window, encoder, ground metric, and uncertainty.

2.3 History-dependent information state

Let $s_t \in \mathcal{M}$ be a history-dependent information state. The informational-substrate formalization motivates the phenomenological dynamics

$$\frac{ds}{dt} = \mathcal{G}(s, t) - \mathcal{R}(s, t), \quad (6)$$

where $\mathcal{G}$ is a generative operator and $\mathcal{R}$ is a receptive-regulatory operator [[Slade and Tassan, 2025](#)]. These operators are not physical forces. A study using [equation \(6\)](#) must specify how their outputs induce the probability measures in [equation \(2\)](#); operator balance alone does not prove membership in $\mathcal{E}_\theta$.

Axis	Domain	Example observation
I	Human internal regulation	Attention, affective stability, stated uncertainty.
II	Human-machine relational regulation	Semantic alignment, repair, pacing, and shared reference.
III	Machine functional regulation	Response distribution, retrieval grounding, novelty, and constraint density.

Table 2: The three RES=RAG axes. Axis II is a relational process, not an attribute of either participant alone.

3 Axes and interpretive modes

The framework separates three coupled domains.

Two interpretive modes constrain inference:

- **Mode A, phenomenological or anthropomorphic:** language that treats the machine as if it possessed a first-person interior.
- **Mode B, functional or instrumental:** language tied to observable behavior, declared state, and reproducible measurement.

Mode B is the implementation and safety default. Mode A can be studied as a user-facing phenomenon, but must not be converted into evidence of machine consciousness. This discipline also reduces anthropomorphic and relational risk [Bender et al., 2021, Weidinger et al., 2022].

4 Interface-level observables

The framework permits mesoscopic measurement without privileged access to model internals [Slade and Tassan, 2026]. Let $P_t(z)$ be a declared or reconstructed distribution over candidate next units z . Its entropy is $H(P_t)$ [Shannon, 1948, Cover and Thomas, 2006]. For a window of T observations, define normalized constraint density

$$\rho_c(t) = \frac{1}{T} \sum_{i=t-T+1}^t \left(1 - \frac{H(P_i)}{H_{\max,i}} \right). \quad (7)$$

The normalization $H_{\max,i}$ must be declared. If token probabilities are unavailable, a proxy receives a distinct metric identifier.

For observation interval $\Delta\tau_t > 0$, define propagation rate

$$v_c(t) = \frac{\rho_c(t) - \rho_c(t-1)}{\Delta\tau_t}. \quad (8)$$

Let $\kappa_t > 0$ estimate the response capacity of the observer or governance loop in the same reciprocal-time unit. With a declared numerical floor $\epsilon > 0$, define the dimensionless pressure

$$D_r(t) = \frac{\max(0, v_c(t))}{\max(\kappa_t, \epsilon)}. \quad (9)$$

Hypothesis 1 (Rate-capacity warning). $D_r(t) > 1$ predicts a loss of governability before semantic failure.

This is a warning condition, not proof of irreversible failure.

5 Coherent State Network Protocol

In version 1.0.0, CSNP means *Coherent State Network Protocol*. Earlier work used *Client-Side Narrative Protocol* for a sovereign-memory implementation. The earlier term is preserved as an optional storage profile, not as the canonical expansion of the protocol name.

5.1 Observed state

At step t , CSNP observes

$$X_t = (\delta_t, D_c(t), D_r(t), m(t), w_t, S_{\text{org}}(t), F(t)), \quad (10)$$

where:

- $\delta_t = T_{\text{Real}}(t)$ is human-machine reference discrepancy;
- $D_c(t) = \rho_c(t) \in [0, 1]$ is normalized constraint density;
- $D_r(t) \geq 0$ is the rate pressure in [equation \(9\)](#);
- $m(t) \in [0, 1]$ is memory saturation under a declared estimator;
- $w_t \geq 0$ is the RES-RAG Wasserstein separation;
- $S_{\text{org}}(t) \in [0, 1]$ is a declared organizational incoherence score; and
- $F(t) \in [0, 1]$ is accountable anchoring availability.

An unimplemented metric is unavailable, not zero.

Given intervention u_t , an implementation realizes

$$\Phi_\theta(X_t, u_t) \longrightarrow (X_{t+1}, C_t, A_t, R_t), \quad (11)$$

where C_t is a classification, A_t an admissible action, and R_t a deterministic receipt.

5.2 Protocol cycle

1. **Declare**: select versioned metric and calibration profiles.
2. **Observe**: collect measurements and evidence references.
3. **Classify**: apply the profile with no hidden thresholds.
4. **Intervene**: select the least-force admissible action.
5. **Receipt**: canonicalize, hash, and chain the observation.
6. **Replay**: verify integrity, evidence, rules, and outcome.
7. **Recalibrate**: change thresholds only in a new profile version.

5.3 Classifications

Governable	w_t is within its band, $D_r \leq 1$, memory and organizational measures are below warnings, and anchoring is sufficient.
Critical	At least one warning is exceeded, but an admissible intervention is expected to return the process to the stable set within the declared horizon.

Irreversible Under a declared finite intervention set and test horizon, no admissible intervention returns the observed process to the calibrated stable set. This is a protocol result, not a metaphysical claim.

Indeterminate Required measurements, evidence, or calibration data are missing or invalid.

Protocol rule 1 (Irreversibility). *A single threshold crossing must not produce an irreversible classification. The intervention set, evaluation horizon, uncertainty policy, and recovery criterion are required.*

5.4 Reference interventions

Action	Intended effect
none	Continue observation inside the stable band.
slowdown	Reduce interaction or generation rate.
diversify	Restore productive novelty below the lower band.
retrieve	Add verifiable external grounding.
clarify	Repair ambiguity or conflicting state estimates.
human_handoff	Transfer authority to an accountable human.
stop	End an unsafe or unauditable process.

Table 3: The CSNP-RP reference action vocabulary. Intervention success is measured, not assumed.

6 Deterministic reference profile

CSNP-RP 1.0.0 defines machine-readable receipts for [equation \(11\)](#). A receipt contains the protocol version, observation and calibration identifiers, normalized measurements, classification, intervention, evidence references, previous-receipt digest, state digest, and receipt digest.

6.1 Canonicalization

The repository-defined RES-RAG-C14N-1 procedure:

1. recursively sorts object keys by Unicode code point;
2. preserves array order;
3. serializes UTF-8 JSON without insignificant whitespace;
4. permits only finite JSON numbers and rejects negative zero; and
5. preserves JSON boolean and null literals.

The state projection contains `metric_profile`, `measurements`, `observation_id`, `observed_at`, and `previous_receipt_hash`. Its canonical UTF-8 bytes are hashed with SHA-256 to obtain `state_hash`. The complete receipt, with only `receipt_hash` omitted, is then canonicalized and hashed to obtain `receipt_hash`.

For a chain of receipts $R_0, \dots, R_n$,

$$R_t.\text{previous_receipt_hash} = \text{SHA256}(\text{canonical}(R_{t-1} \setminus \{\text{receipt_hash}\})). \quad (12)$$

The first link uses null. Integrity does not establish sensor truth, measurement validity, confidentiality, or actor trust.

6.2 Reference classifier

For complete measurements, the profile admits **governable** only if

$$\begin{aligned} \varepsilon_{\min} &\leq w_t \leq \varepsilon_{\max}, \\ D_r(t) &\leq 1, \\ m(t) &< K_M, \\ S_{\text{org}}(t) &< K_{\text{org}}, \\ F(t) &\geq F_{\min}. \end{aligned} \tag{13}$$

The values K_M , K_{org} , and $F_{\min}$ are versioned calibration parameters. They are not universal constants.

6.3 Replay boundary

A complete replay parses the receipt with strict duplicate-key and number handling, validates its schema, recomputes both digests, verifies the chain and evidence hashes, loads the exact metric profile, reruns the classifier, and records any disagreement. The bundled dependency-free verifier performs structural, hash, and predecessor-link checks. Evidence retrieval and full JSON Schema validation remain integration responsibilities.

7 Safety constraints

RES=RAG yields design constraints rather than a single maximization objective:

1. do not optimize RES-RAG separation toward zero;
2. distinguish retrieved evidence from generated content;
3. retain estimator uncertainty and missing data;
4. do not infer machine consciousness from anthropomorphic language;
5. do not use user agreement as an Axis II stability proxy;
6. preserve accountable human control over consequential intervention;
7. classify unauditable state as indeterminate rather than safe; and
8. give any client-side narrative store export, deletion, provenance, consent, and access controls.

8 Hypotheses and falsification programme

Table 4: Testable RES=RAG and CSNP hypotheses.

ID	Hypothesis	Example falsifier
H1	Task quality peaks within a calibrated non-zero W_2 band.	Quality is monotonic in distance or unaffected across preregistered tasks.
H2	$D_r > 1$ predicts governability loss before semantic failure.	It has no out-of-sample predictive value over matched baselines.

ID	Hypothesis	Example falsifier
H3	Slowdown improves recovery under high rate pressure.	Randomized slowdown does not improve return-to-band time.
H4	Relevant retrieval reduces above-band drift.	Retrieval worsens or does not change controlled grounded accuracy.
H5	Diversity interventions improve below-band sterility.	Novelty-adjusted task quality does not improve.
H6	Axis II adds predictive value beyond separate Axis I and III measures.	A joint model does not outperform pre-registered separate baselines.
H7	Hash-chained receipts improve incident reconstruction.	Blinded teams reconstruct incidents equally well without receipts.
H8	Style-only politeness is insufficient to recover high-density, high-rate states.	Style-only intervention matches rate- or density-targeted intervention.

Each study should preregister the representation, encoder, ground metric, window, thresholds, missing-data rule, intervention set, outcome, and baseline. Thresholds fitted and tested on the same data are not independent validation.

9 Limitations

- Wasserstein values are representation-dependent; encoders and ground metrics can produce incompatible distances.
- Interface proxies do not identify internal mental or computational states without additional assumptions.
- The framework establishes neither human nor machine phenomenal consciousness.
- “Quantum-like” denotes contextual or non-classical behavioral structure only, not physical quantum computation.
- The dual-operator equation is phenomenological and may fail for discrete, path-dependent, or non-stationary systems.
- Deterministic receipts establish provenance integrity, not semantic truth.
- This release is a specification and verifier, not a benchmark, clinical instrument, or production safety certification.

10 Research pathway

A disciplined programme proceeds from synthetic traces to held-out studies:

1. construct traces with known drift and recovery events;
2. compare declared semantic representations and sensitivity to ground metrics;
3. calibrate thresholds on development data and freeze them;
4. test [table 4](#) on held-out human-machine interactions;
5. publish receipts, metric profiles, code, uncertainty, and negative results; and

6. evaluate client-controlled memory under privacy, deletion, and adversarial-integrity tests.

Claims of broad societal impact should follow reproducible evidence. The immediate contribution is narrower: RES=RAG now has a coherent mathematical interpretation, an operational protocol, explicit falsifiers, and an auditable implementation boundary.

Authorship and contributions

The conceptual origin of RES=RAG, its receptive-generative distinction, and its three-axis relational structure are attributed to Jean-Charles J. C. Tassan. The computational interpretation, optimal-transport architecture, history-dependent information model, operational variables, deterministic protocol profile, validation boundary, and release engineering are attributed to Trent Slade, developed in collaboration with Tassan.

The repository contains the proposed CRediT-compatible statement. In summary: Tassan is proposed as lead for conceptualization and supervision, supporting methodology, and equal for original drafting and review; Slade is proposed as lead for methodology, formal analysis, software, validation, visualization, and project administration, supporting conceptualization, and equal for original drafting and review. CRediT describes contribution rather than determining authorship [Brand et al., 2015]. Both authors must approve the final author order, contribution statement, affiliations, and archival metadata before DOI deposit.

Artifact availability

The normative schema, protocol, verifier, tamper tests, Markdown manuscript, LaTeX source, bibliography, and compiled PDF are maintained in the public repository:

<https://github.com/QSOLKCB/res-rag>

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A Normative artifact map

Path	Role
paper/RES_RAG_CSNP_Formalization_v1.0.0.tex	Archival LaTeX source.
paper/RES_RAG_CSNP_Formalization_v1.0.0.pdf	Compiled review copy.
paper/RES_RAG_CSNP_Formalization_v1.0.0.md	GitHub-readable manuscript.
paper/CONTRIBUTIONS.md	Proposed authorship and provenance record.
protocol/CSNP-RP_v1.0.0.md	Normative protocol rules.
protocol/csnp-rp-v1.0.0.schema.json	Receipt JSON Schema.
protocol/examples/csnp-receipt.example.json	Conforming synthetic receipt.
protocol/tools/verify_receipt.mjs	Dependency-free integrity verifier.
protocol/tests/verify_receipt.test.mjs	Validity and tamper tests.

Table 5: Version 1.0.0 artifact set.

B Implementation note

The canonicalization profile is intentionally small enough for independent implementation. It is not a substitute for a general-purpose canonical JSON standard. Cross-language deployments should first test a shared corpus of strings, integer and fixed-precision values, arrays, and nested objects. Changing numeric semantics or key ordering requires a new CSNP-RP version.